Theorem 1: blowup distances equal optimal transport distances

Correspondence between the paper and the Lean 4/Mathlib formalisation

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport* (file `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository). Statement, equation, page and line numbers refer to this PDF; `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

The folder `GsimOT/Theorem1_BlowupEqualsOT` formalises Theorem 1 and its proof in Appendix B, including Claim 20. The operator norms, Lemma 19 and the attainment of the minimum in Eq. (2) live in `GsimOT/Common` (`OperatorNorm.lean`, `Attainment.lean`). A proof overview is `Theorem1_overview.pdf`.

Checking. Run `bash build.sh` in the repository root (see `README.md`). The script builds the library and runs `scripts/CheckAxioms.lean`, which must report only the standard axioms `propext`, `Classical.choice`, `Quot.sound`: no `sorry` and no additional axioms.

Main statement in Lean. `GsimOT.theorem1`: for all $p, q \in [1, \infty]$ and all finite graphs G on $V \neq \emptyset$, H on $W \neq \emptyset$,

$$\delta^{\bullet}_{p\to q}(G, H) = \delta^{\mathrm{OT}}_{p\to q}(G, H).$$

`GsimOT.theorem1_tendsto`: moreover $\delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell}) \to \delta^{\mathrm{OT}}_{p\to q}(G, H)$ as $\ell \to \infty$, where $m = |V|$, $n = |W|$ (so the limit exists). `GsimOT.theorem1_of_tendsto`: the paper's argument, assuming the existence of the limit.

Correspondence: Theorem 1 and its proof

Paper	Page	Statement	Lean declaration(s) and location
Theorem 1	3	$\delta^{\bullet}_{p\to q} = \delta^{\mathrm{OT}}_{p\to q}$ for all $1 \le p, q \le \infty$	`GsimOT.theorem1` `Theorem1_BlowupEqualsOT/Theorem1.lean:68`
Theorem 1 (strengthened)	3	$\delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell}) \to \delta^{\mathrm{OT}}_{p\to q}(G, H)$ (the limit exists)	`GsimOT.theorem1_tendsto` `Theorem1_BlowupEqualsOT/Theorem1.lean:48`
Theorem 1 (paper's proof)	3, 15–20	If the limit defining $\delta^{\bullet}_{p\to q}$ exists (cited from [Gro25], p. 2), it equals $\delta^{\mathrm{OT}}_{p\to q}$; upper bound along $\ell = ks$ only	`GsimOT.theorem1_of_tendsto` `Theorem1_BlowupEqualsOT/Claim20.lean:151`

Section 1	1–2	Blowup $G^{\bullet k}$ and blowup distance $\delta^{\bullet} = \lim_{\ell} \delta(G^{\bullet n\ell}, H^{\bullet m\ell})$	`GsimOT.blowup`
			`Theorem1_BlowupEqualsOT/Definitions.lean:29`
			`GsimOT.blowupSeq`
			`Theorem1_BlowupEqualsOT/Definitions.lean:47`
			`GsimOT.deltaBlowup`
			`Theorem1_BlowupEqualsOT/Definitions.lean:51`

Paper	Page	Statement	Lean declaration(s) and location
Section 1	2	Fractional alignments $\mathrm{DS}^*(V,W)$	GsimOT.IsCoupling Common/OTDistances.lean:36 GsimOT.isCoupling_uniform Theorem1_BlowupEqualsOT/Theorem1.lean:34
Section 1, Appendix A, B	3, 14, 15	$B_{p\to q}(G,H,Q^*)$ with $0^0 := 0$; $\kappa_{p\to q} = 1/q + 1 - 1/p$	GsimOT.powZ Common/OTDistances.lean:42 GsimOT.Bmat Common/OTDistances.lean:70 GsimOT.kappa Common/AlignedDistances.lean:23
Eq. (2)	3	$\delta^{\mathrm{OT}}_{p\to q}(G,H) = \min_{Q^*} \|B_{p\to q}(G,H,Q^*)\|_{p\to q}$ (infimum)	GsimOT.deltaOTPQ Common/OTDistances.lean:74
Appendix A	14–15	The minimum in Eq. (2) is attained (lower semicontinuity, compactness of $\mathrm{DS}^*(V,W)$)	GsimOT.lowerSemicontinuous_opNorm_Bmat Common/Attainment.lean:136 GsimOT.isCompact_setOf_isCoupling Common/Attainment.lean:168 GsimOT.exists_deltaOTPQ_eq Common/Attainment.lean:199
Section 1, 2, Appendix B	1, 3, 5, 20	A^π_G, $d_{p\to q} = \min_\pi \|A^\pi_G - A_H\|_{p\to q}$, $\delta_{p\to q} = N^{-\kappa} d_{p\to q}$	GsimOT.alignedDiff Common/AlignedDistances.lean:31 GsimOT.distPQ Common/AlignedDistances.lean:35 GsimOT.deltaPQ Common/AlignedDistances.lean:39
Proof of Eq. (9)	15–16	$J(v,w)$ as blocks of the map $(w,j) \mapsto (v(w,j),w)$; $Q^*(v,w) = \|J(v,w)\|/N$	GsimOT.blockCard Theorem1_BlowupEqualsOT/BlockNorms.lean:24 GsimOT.blockMap Theorem1_BlowupEqualsOT/LowerBound.lean:50 GsimOT.couplingOf Theorem1_BlowupEqualsOT/LowerBound.lean:53

Paper	Page	Statement	Lean declaration(s) and location
Proof of Eq. (9)	15	$Q^* \in \mathrm{DS}^*(V, W)$ (row sums $\ell n/N$, column sums $\ell m/N$)	`GsimOT.sum_blockCard_row` Theorem1_BlowupEqualsOT/LowerBound.lean:57 `GsimOT.sum_blockCard_col` Theorem1_BlowupEqualsOT/LowerBound.lean:76 `GsimOT.isCoupling_couplingOf` Theorem1_BlowupEqualsOT/LowerBound.lean:96
Eq. (10)	15	$M_\pi((w,j),(w',j')) = D(v,w,v',w')$ for $j \in J(v,w)$, $j' \in J(v',w')$	`GsimOT.alignedDiff_blowup_eq_disc` Theorem1_BlowupEqualsOT/LowerBound.lean:117
Proof of Eq. (11)	16	Spread test vectors: $\|\tilde{x}\|_p, \|\tilde{y}\|_{q^*} \le 1$	`GsimOT.lpNorm_spread_le` Theorem1_BlowupEqualsOT/BlockNorms.lean:40
Proof of Eq. (11)	17	Chain of equalities $\tilde{y}^\top M_\pi \tilde{x} = N^\kappa\, y^\top B x$	`GsimOT.sum_comp_eq_sum_blockCard` Theorem1_BlowupEqualsOT/LowerBound.lean:20 `GsimOT.div_rpow_self_eq_powZ` Theorem1_BlowupEqualsOT/LowerBound.lean:27 `GsimOT.powZ_eq_rpow_mul_powZ_div` Theorem1_BlowupEqualsOT/LowerBound.lean:34
Eq. (11)	15	$\|M_\pi\|_{p\to q} \ge N^\kappa \|B_{p\to q}(G,H,Q^*)\|_{p\to q}$	`GsimOT.eq11` Theorem1_BlowupEqualsOT/LowerBound.lean:127
Eqs. (8), (9)	15	$\delta^{\mathrm{OT}}_{p\to q}(G,H) \le \delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell})$ for all $\ell \ge 1$	`GsimOT.deltaOTPQ_le_blowupSeq` Theorem1_BlowupEqualsOT/LowerBound.lean:204
Claim 20 (proof)	17	Rational couplings Q_k^* with $\mathrm{supp}(Q_k^*) = \mathrm{supp}(Q_0^*)$ and $Q_k^* \to Q_0^*$ (here: denominators ℓmn, by Birkhoff rounding)	`GsimOT.exists_integral_approx_close` Theorem1_BlowupEqualsOT/Approximation.lean:319 `GsimOT.exists_rational_seq_supp_eq` Theorem1_BlowupEqualsOT/Claim20.lean:58

Paper	Page	Statement	Lean declaration(s) and location

(not in paper)	–	Lift Q to an $mn \times mn$ doubly stochastic matrix, Birkhoff, average back	`GsimOT.liftM`
			Theorem1_BlowupEqualsOT/Approximation.lean:217
			`GsimOT.liftM_mem_doublyStochastic`
			Theorem1_BlowupEqualsOT/Approximation.lean:278
			`GsimOT.avgM`
			Theorem1_BlowupEqualsOT/Approximation.lean:212
			`GsimOT.isCoupling_avgM`
			Theorem1_BlowupEqualsOT/Approximation.lean:222
			`GsimOT.avgM_liftM`
			Theorem1_BlowupEqualsOT/Approximation.lean:295
			`GsimOT.permMatrix_apply_mem_rangeS`
			Theorem1_BlowupEqualsOT/Approximation.lean:306

(not in paper)	–	Rounding convex weights to multiples of $1/\ell$ with the same support	`GsimOT.exists_round_weights` Theorem1_BlowupEqualsOT/Approximation.lean:105

Claim 20 (proof)	17	On a fixed support, $\|B_{p\to q}(G,H,Q_k^*)\| \to \|B_{p\to q}(G,H,Q_0^*)\|$	`GsimOT.tendsto_powZ_apply`
			Theorem1_BlowupEqualsOT/Approximation.lean:45
			`GsimOT.exists_delta_opNorm_Bmat`
			Theorem1_BlowupEqualsOT/Approximation.lean:68
			`GsimOT.tendsto_opNorm_Bmat_of_supp_eq`
			Theorem1_BlowupEqualsOT/Claim20.lean:40

Claim 20, Eq. (13)	17	A rational $Q^* \in \mathrm{DS}^*(V,W)$ with $\|B_{p\to q}(G,H,Q^*)\| \le \min_{Q_0^*} \|B_{p\to q}(G,H,Q_0^*)\| + \varepsilon$	`GsimOT.claim20` Theorem1_BlowupEqualsOT/Claim20.lean:92
Claim 20 (strengthened)	17	Near-optimal couplings with denominator $N = \ell mn$ for *every* large ℓ	`GsimOT.exists_integral_approx` Theorem1_BlowupEqualsOT/Approximation.lean:379

Proof of Eq. (15)	18	Partitions $I(v,w)$, $J(v,w)$ of sizes $q(v,w) = NQ^*(v,w)$ and the bijection π	`GsimOT.exists_fin_fun_card_fiber`
			Theorem1_BlowupEqualsOT/UpperBound.lean:26
			`GsimOT.card_fiber_prod`
			Theorem1_BlowupEqualsOT/UpperBound.lean:37
			`GsimOT.exists_blowup_bij`
			Theorem1_BlowupEqualsOT/UpperBound.lean:49

Proof of Eq. (15)	18–19	Averaged vectors: $\|x\|_p, \|y\|_{q^*} \le 1$ (Hölder, Eq. (5))	`GsimOT.lpNorm_average_le` Theorem1_BlowupEqualsOT/BlockNorms.lean:73

Paper	Page	Statement	Lean declaration(s) and location
Proof of Eq. (15)	19–20	Chain of equalities $\tilde{y}^\top M_\pi \tilde{x} = N^\kappa\, y^\top B x$	`GsimOT.block_sum_eq` `Theorem1_BlowupEqualsOT/UpperBound.lean:78`
Eq. (15)	18	$N \cdot Q^*$ integral $\Rightarrow \delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell}) \le \|B_{p\to q}(G, H, Q^*)\|_{p\to q}$	`GsimOT.eq15_of_integral` `Theorem1_BlowupEqualsOT/UpperBound.lean:99`
Eq. (15), $\ell = ks$	18	k clears the denominators of Q^* $\Rightarrow \delta_{p\to q}(G^{\bullet nks}, H^{\bullet mks}) \le \|B_{p\to q}(G, H, Q^*)\|$ for all $s \ge 1$	`GsimOT.exists_denom` `Theorem1_BlowupEqualsOT/Claim20.lean:106` `GsimOT.eq15` `Theorem1_BlowupEqualsOT/Claim20.lean:134`
Eqs. (12), (14)	17–18	Conclusion: $\delta^\bullet \le \delta^{\mathrm{OT}}$, hence equality	`GsimOT.theorem1_tendsto` `Theorem1_BlowupEqualsOT/Theorem1.lean:48` `GsimOT.theorem1_of_tendsto` `Theorem1_BlowupEqualsOT/Claim20.lean:151`

Operator norms and Lemma 19 used (from `GsimOT/Common`)

Paper	Page	Statement	Lean declaration(s) and location
Appendix A	14	ℓ^p norm $\|x\|_p$ $(1 \le p \le \infty)$	`GsimOT.lpNorm` `Common/OperatorNorm.lean:25` `GsimOT.lpNorm_eq_sum` `Common/OperatorNorm.lean:57` `GsimOT.lpNorm_top` `Common/OperatorNorm.lean:62`
Appendix A	14	Operator norm $\|A\|_{p\to q} = \max_{\|x\|_p=1} \|Ax\|_q$ (in Lean the supremum over the unit ball $\|x\|_p \le 1$, the same value for nonempty index sets)	`GsimOT.opNorm` `Common/OperatorNorm.lean:28` `GsimOT.lpNorm_mulVec_le` `Common/OperatorNorm.lean:128` `GsimOT.opNorm_le_of_bound` `Common/OperatorNorm.lean:135` `GsimOT.exists_lpNorm_mulVec_eq_opNorm` `Common/OperatorNorm.lean:142`

Paper	Page	Statement	Lean declaration(s) and location
Appendix A	14	Hölder conjugate q^* with $1/q^* = 1 - 1/q$	`GsimOT.one_le_conjExponent` Common/OperatorNorm.lean:33 `GsimOT.inv_conjExponent_toReal` Common/OperatorNorm.lean:38
Appendix A	14	Hölder's inequality $\lvert y^\top z\rvert \le \lVert y\rVert_{q^*}\lVert z\rVert_q$ and its equality case	`GsimOT.abs_dotProduct_le_lpNorm_mul` Common/OperatorNorm.lean:188 `GsimOT.exists_dotProduct_eq_lpNorm` Common/OperatorNorm.lean:224
Lemma 19	14	$\lVert A\rVert_{p\to q} = \max_{\lVert y\rVert_{q^*}=1,\lVert x\rVert_p=1} \lvert y^\top Ax\rvert$ (in Lean over the unit balls)	`GsimOT.lemma19` Common/OperatorNorm.lean:290 `GsimOT.abs_dotProduct_mulVec_le_opNorm` Common/OperatorNorm.lean:267 `GsimOT.exists_abs_dotProduct_mulVec_eq_opNorm` Common/OperatorNorm.lean:281
–	–	Triangle inequality, $\lVert A\rVert_{p\to q} \le \sum_{i,j} \lvert A_{ij}\rvert$ and continuity of $\lVert\cdot\rVert_{p\to q}$	`GsimOT.opNorm_add_le` Common/OperatorNorm.lean:169 `GsimOT.opNorm_le_sum_abs` Common/OperatorNorm.lean:158 `GsimOT.abs_opNorm_sub_le` Common/Attainment.lean:78 `GsimOT.continuous_opNorm` Common/Attainment.lean:91
Appendix A	15	Restricting to rows and columns "cannot increase the operator norm"	`GsimOT.lpNorm_le_of_abs_le` Common/Attainment.lean:39 `GsimOT.opNorm_restrict_le` Common/Attainment.lean:53

Notes on the formalisation

1. **Two versions of the final step.** The paper takes the existence of the limit $\delta^\bullet = \lim_\ell \delta_{p\to q}(G^{\bullet n\ell}, H^{\bullet m\ell})$ from [Gro25] (p. 2) and proves the upper bound only along the subsequence $\ell = ks$ (Eq. (15)). `theorem1_of_tendsto` is this argument verbatim, with the existence of the limit as a hypothesis. The main theorem `theorem1` needs no such hypothesis: its approximation step (`exists_integral_approx`) works for *every* sufficiently large ℓ, so the whole sequence converges to $\delta^{\mathrm{OT}}_{p\to q}$ (`theorem1_tendsto`). In Lean, $\delta^\bullet$ is `limUnder` of the sequence (`deltaBlowup`).

2. **Claim 20** (p. 17) is formalised as `claim20` and follows the paper: a minimiser Q_0^* (`exists_deltaOTPQ_eq`, from the attainment argument of Appendix A), rational couplings $Q_k^* \to Q_0^*$ with the same support, and convergence of the norms on a fixed support. The rational sequence is obtained differently: the paper uses that the couplings vanishing outside S form a rational polytope; we construct the sequence explicitly. Q_0^* is lifted to the doubly stochastic $mn \times mn$ matrix $R((v,i),(w,j)) = Q_0^*(v,w)$, decomposed by Birkhoff–von Neumann (Mathlib), the positive coefficients are rounded to positive multiples of $1/\ell$, and the result is averaged back. This preserves the support, which is essential: with $0^0 = 0$, $Q \mapsto B_{p\to q}(Q)$ is discontinuous at the boundary of the support for $p = 1$ or $q = \infty$.

3. Eqs. (11) and (15) follow the paper's argument verbatim: the coupling $Q^* = |J|/N$, Eq. (10), maximising test vectors from Lemma 19, the spread (resp. averaged) vectors and the chain of equalities. The norm computations for these vectors are factored out into the general block lemmas `lpNorm_spread_le` and `lpNorm_average_le`, for a block map $b : \beta \to \alpha$ with blocks $\{w\} \times J(v,w)$. `eq11` is Eq. (11) and `eq15_of_integral` is Eq. (15).

4. **Normalisation.** $\delta_{p\to q} = N^{-\kappa_{p\to q}} d_{p\to q}$ for graphs of order N, i.e. the metric is scaled by $1/s(N)$ with $s(N) = N^{1/q+1-1/p}$ (p. 3, l. 88–90; Appendix B, p. 15, l. 513–514, and p. 20, l. 637–638).

5. Operator norms are defined as the supremum of $\|Ax\|_q$ over the unit *ball* $\|x\|_p \le 1$, and Lemma 19 is stated with $\|x\|_p \le 1$, $\|y\|_{q^*} \le 1$. For nonempty index sets this equals the paper's maximum over unit spheres by homogeneity; the paper's proof notes this itself (p. 20, l. 636–637).

6. Theorem 1 is formalised for general $p, q \in [1,\infty]$ (`ENNReal`). It is independent of the local edit distance development: for $p = q = 1$ it concerns $\|B_{1\to 1}\|_1$, which the paper relates to C_1 in Eq. (7); this link is proved in Common (`dOT_one_one_eq_dOTOne`) but not needed here.